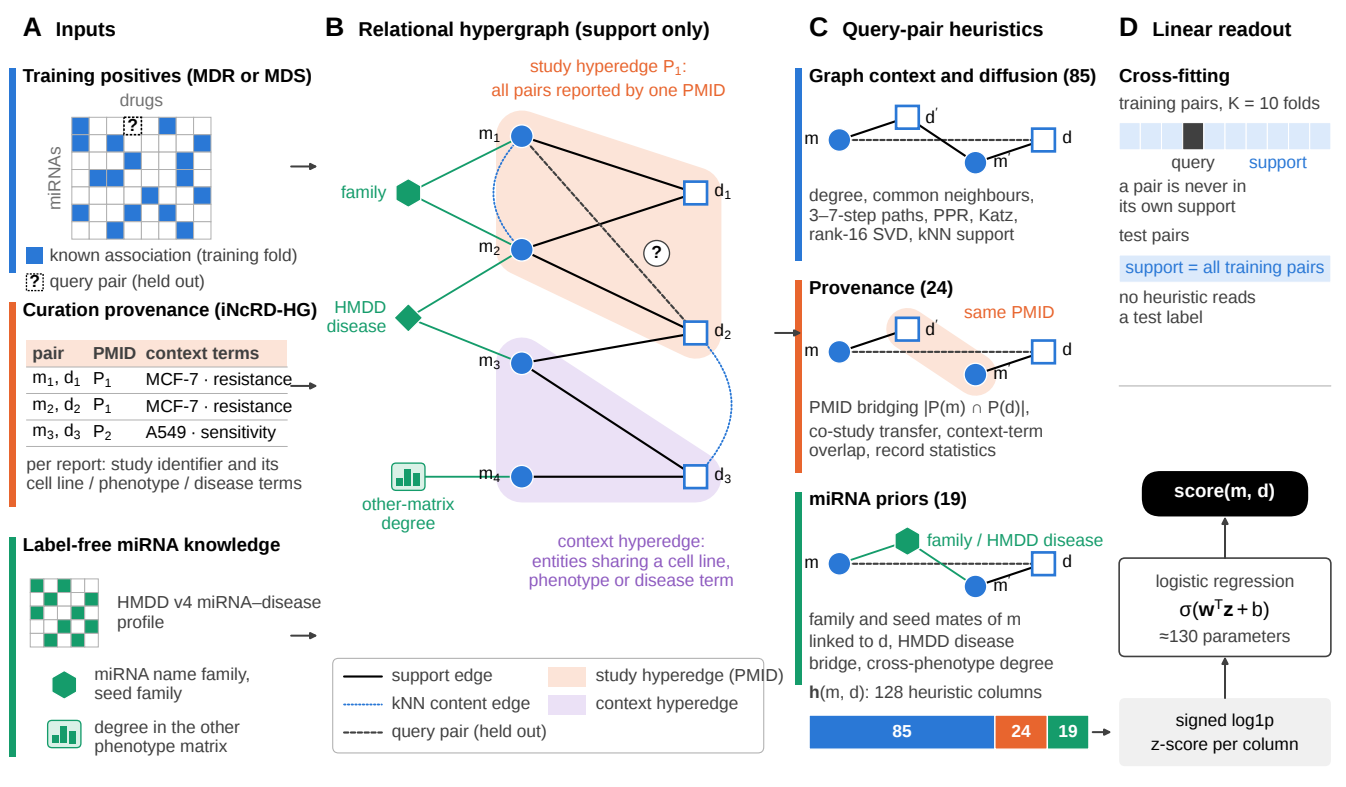
A Inputs
Training positives (MDR or MDS)
drugs
miRNAs
known association (training fold)
query pair (held out)
Curation provenance (iNcRD-HG)
pair PMID context terms
m1, d1 P1 MCF-7 · resistance
m2, d2 P1 MCF-7 · resistance
m3, d3 P2 A549 · sensitivity
per report: study identifier and its cell line / phenotype / disease terms
Label-free miRNA knowledge
HMDD v4 miRNA–disease profile
miRNA name family, seed family
degree in the other phenotype matrix
B Relational hypergraph (support only)
study hyperedge P1: all pairs reported by one PMID
family
HMDD disease
other-matrix degree
context hyperedge: entities sharing a cell line, phenotype or disease term
support edge
kNN content edge
query pair (held out)
study hyperedge (PMID)
context hyperedge
C Query-pair heuristics
Graph context and diffusion (85)
degree, common neighbours, 3–7-step paths, PPR, Katz, rank-16 SVD, kNN support
Provenance (24)
same PMID
PMID bridging |P(m) ∩ P(d)|, co-study transfer, context-term overlap, record statistics
miRNA priors (19)
family / HMDD disease
family and seed mates of m linked to d, HMDD disease bridge, cross-phenotype degree
h(m, d): 128 heuristic columns
85
24
19
D Linear readout
Cross-fitting
training pairs, K = 10 folds
query
support
a pair is never in its own support
test pairs
support = all training pairs
no heuristic reads a test label
score(m, d)
logistic regression
σ(wᵀz + b)
≈130 parameters
signed log1p z-score per column